MIL-M-28787/135A(EC)
27 March 1978
SUPERSEDING
MIL-M-28787/135(NAVY)
13 May 1974

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INDUCTOR, 1.5 AMPERE, POWER, KEY CODE QPM

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of a power module which contains five identical inductors.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Inductor, 1.5 ampere
Family	Power
Key code	QPM
Size	1D
Weight	350 grams maximum
Failure rate	0.283 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	1.3 W
Rated input voltage	2.12 Vrms, 2,400 Hz
Insulation voltage rating	200 Vdc minimum
Continuous operating conditions	2 Adc, 6V p-p, 2,400 Hz 2 Adc, 2V p-p, 4,800 Hz
Inductance	0.575 mH maximum

1.3 Module function. The 1.5 ampere inductor module, designated as key code QPM, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains five identical inductors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D. C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389- Design Requirements for Standard Electronic Modules.

MIL-STD-1665- Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.283 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QPM.

3.5.2 Size. The module size shall be 1D.

3.5.3 Weight. The module shall have a maximum weight of 350 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

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3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Hold-down holes. The module shall have no additional requirements for location or quantity of hold-down holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/135, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The functional diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

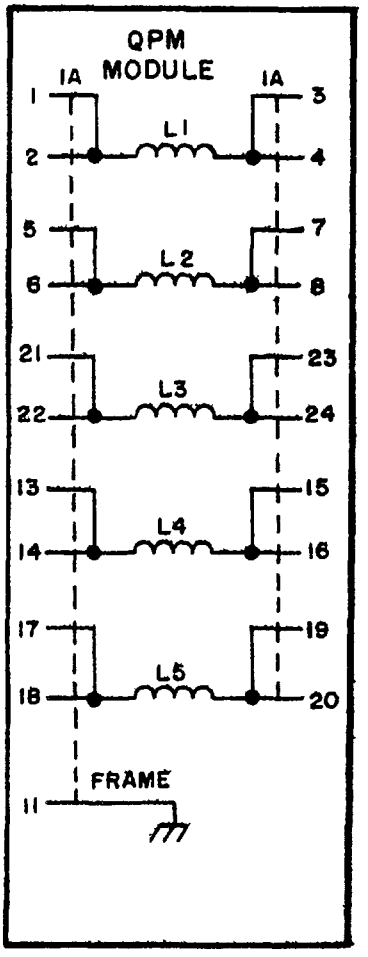
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 Designator	No. required (maximum per test figure)
Digital multifunction meter	M02	2
Power amplifier	A01	1
Sine wave generator	G01	1
Power supply	P01	1
Inductance bridge	L01	1
Megohm bridge	R06	1
Resistors		
0.1 Ω ± 1 % 5W (noninductive)		1
100 Ω ± 1 % 1/8 W		1
Capacitors		
10,000 µF, 10 Vdc (polarized)		1
Inductor		
50 mH or greater at 1.5A (triad C48U or equivalent)		1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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NOTE: L1-L5 = 0.5125 mH ±12.2%.

FIGURE 1. QPM module functional diagram, 1.5 ampere indicator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A10	Continuity	II	X	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
B1-B5	Inductance	III	X	0.45	0.575	0.45	0.575	0.45	0.575	0.45	0.575	mH
C1-C5	DC resistance	IV	X	60.3	73.7	60.3	73.7	54.5	89.6	54.5	89.6	mΩ
D1-D5	Insulation resistance at 200 ±10 Vdc	V	X	1000	∞	1000	∞	1000	∞	1000	∞	MΩ
E	Module integrity at 200 ±10 Vdc	VI	X	1000	∞	1000	∞	1000	∞	1000	∞	MΩ
F	Frame continuity	VII	S	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
G1-G10	Electromagnetic coupling	VIII	S	0.0	40	0.0	40	0.0	40	0.0	40	mV
H1-H10	Jumper resistance	IX	S	0.0	0.020	0.0	0.025	0.0	0.025	0.0	0.030	Ω

NOTES: 1. All (X) parameters shall be 100 percent tested during module acceptance tests.
2. All (S) parameters shall be 25°C ambient sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater).

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through H shall be performed in any order with the following exceptions: If test conditions D and E are part of a sequence of tests, test condition D and E shall be the first tests run. Test conditions A, B, and C, and F, G, and H shall be performed following test conditions D and E.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through H in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through E, G, and H shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column, and test conditions E and F shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, electrical tests A through E, G, and H shall be performed in the order specified in 4.4.2 using the 0° to 60°C operating temperature end-of-life requirements of table I After the module has been returned to 25°C, electrical tests A through H shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

4.5.4 Miscellaneous tests. The following tests shall be performed in accordance with MIL-M-28787: (a) thermal shock, (b) shock, (c) durability, (d) humidity, (e) salt fog, and (f) fungus. No electrical tests shall be performed during these tests but at the conclusion of each environmental test, the module shall conform to the requirements of test conditions E and F. At the completion of all tests, the module shall conform to the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

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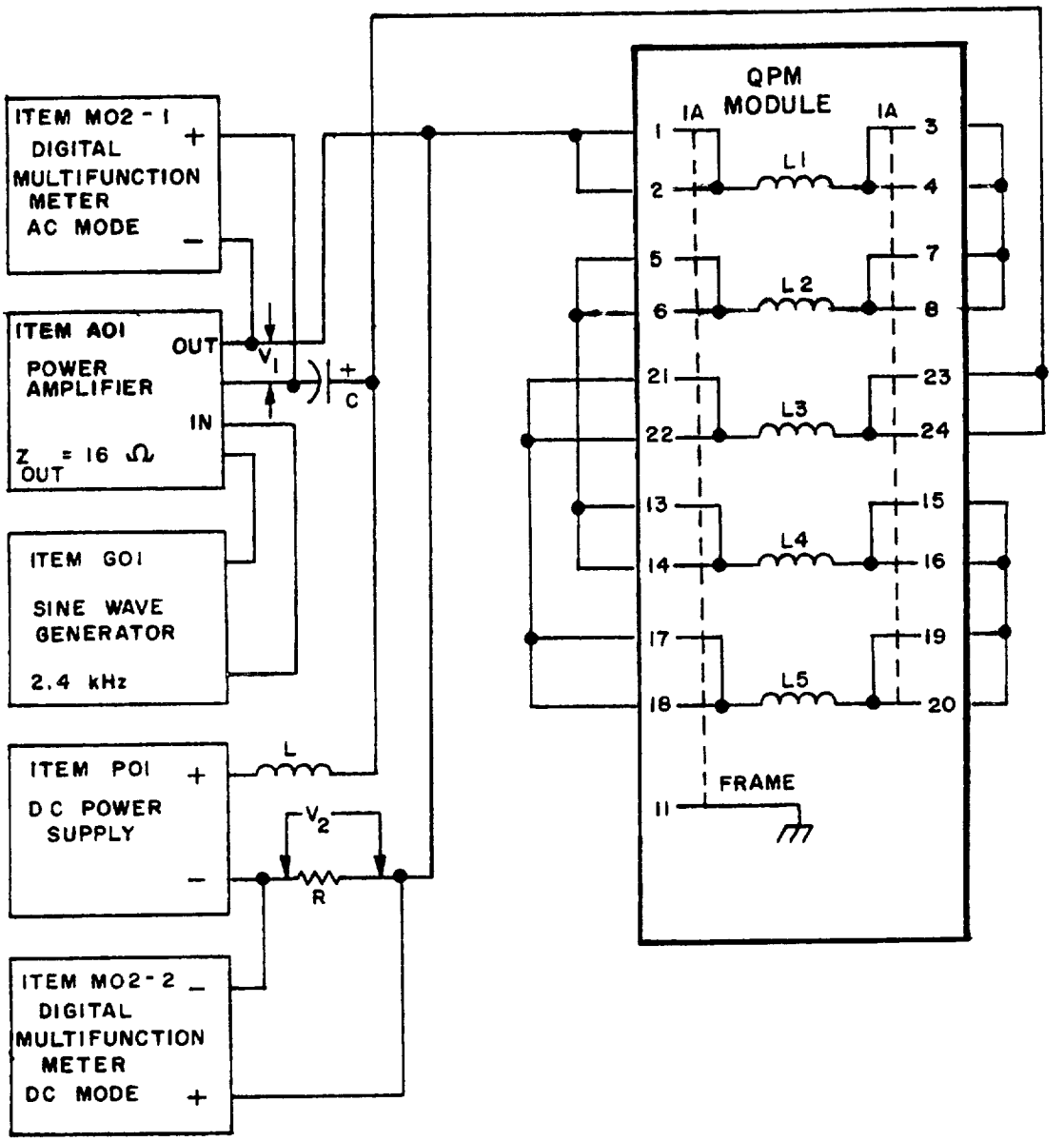


FIGURE 2. Operating temperature and life test circuit.

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NOTES: 1. V_1 = 10.6V rms ±3%
2. V_2 = 0.15V dc ±3%
3. C = 10,000 µF, 10V dc (POL)
4. L = ≥ 50 mH at 1.5A dc
5. R = 0.10 ohm ±1%, 5 W
6. Adjust equipment to obtain voltage of Notes 1 and 2.

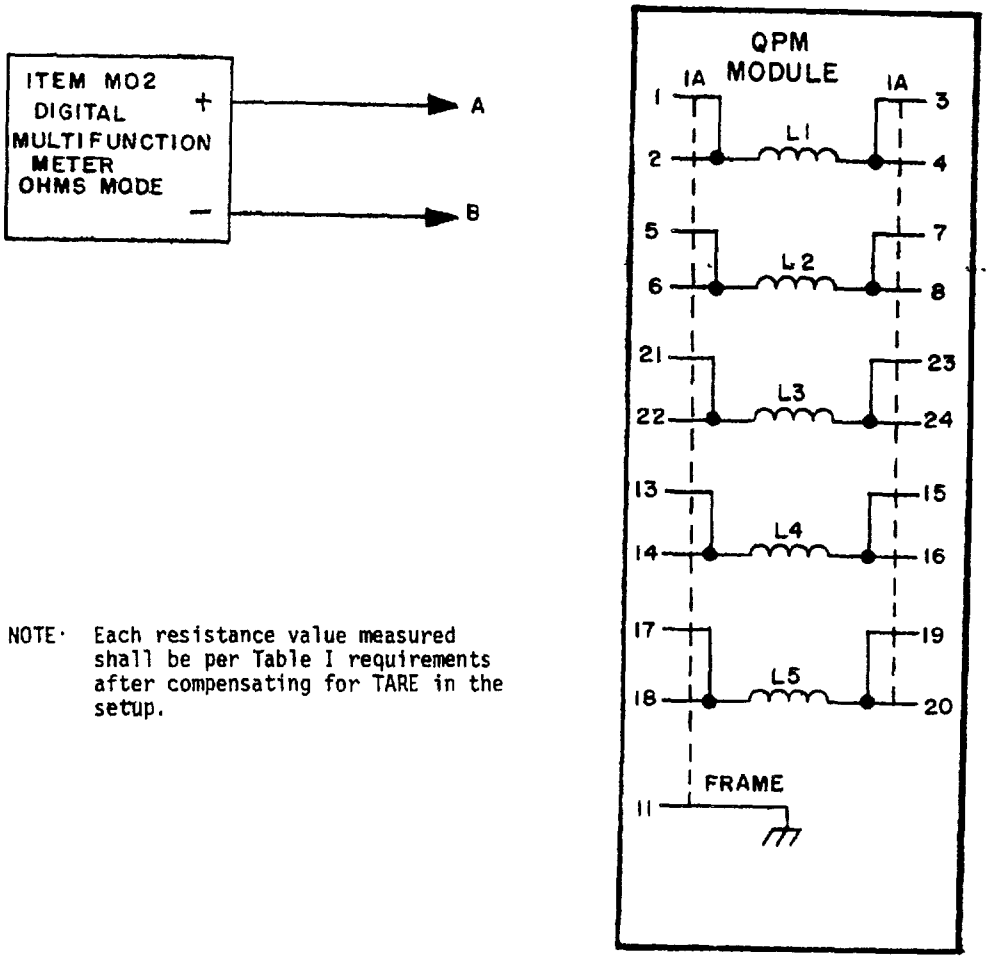
FIGURE 2. Operating temperature and life test circuit - Continued.

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TABLE II. Continuity tests.

Test cond	Parameter	Connect		Initial limits 25° ±5°C amb		Units	Test fig
		A to pin	B to pin	Min	Max		
A1	L1 Continuity	1	2	0.0	0.1	Ω	3
A2	L1	3	4				
A3	L2	5	6				
A4	L2	7	8				
A5	L3	21	22				
A6	L3	23	24				
A7	L4	13	14				
A8	L4	15	16				
A9	L5	17	18				
A10	L5 Continuity	19	20	0.0	0.1	Ω	3

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NOTE: Each resistance value measured shall be per Table I requirements after compensating for TARE in the setup.

FIGURE 3. Continuity, DC resistance, frame continuity, and jumper resistance test circuit for tables II, IV, VII, and IX.

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TABLE III. Inductance tests.

TEST COND	PARAMETER	CONNECT A TO PIN	CONNECT B TO PIN	INITIAL LIMITS 25° ±5°C AMB MIN	INITIAL LIMITS 25° ±5°C AMB MAX	UNITS	TEST FIG
B1	L1 Inductance	1,2	3,4	0.45	0.575	mH	4
B2	L2	5,6	7,8	↓	↓	↓	↓
B3	L3	21,22	23,24	↓	↓	↓	↓
B4	L4	13,14	15,16	↓	↓	↓	↓
B5	L5 Inductance	17,18	19,20	0.45	0.575	mH	4

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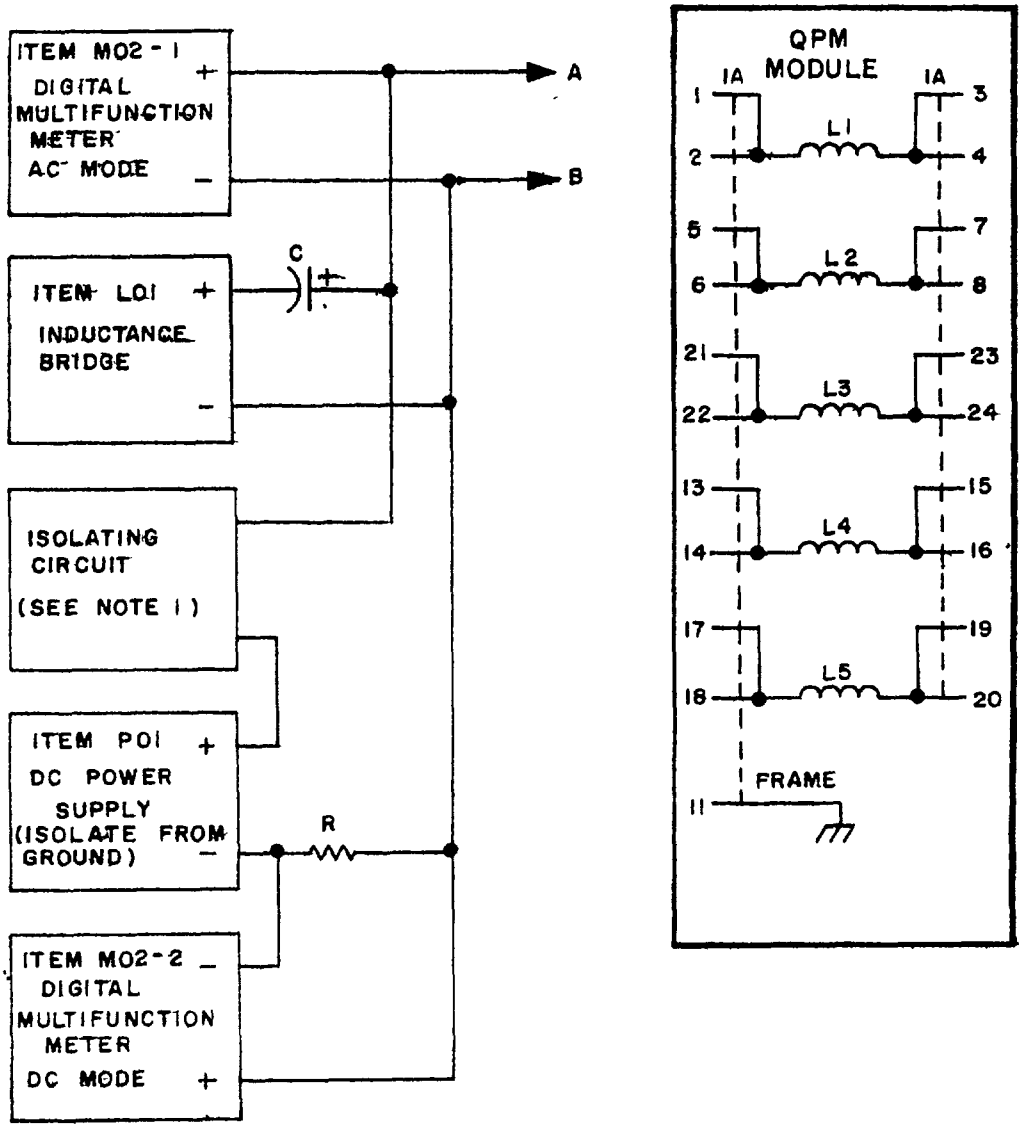


FIGURE 4. Inductance test circuit for table III.

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NOTES:
1. Consists of 50 mH or larger at 1.5A dc choke or transistor insolating circuit (Triad C48U or equivalent).
2. C = 10,000 µF, 10 V, if required.
3. R = .1 ohm ±1%, 5 W.
4. Inductance bridge test voltage shall be 2.00 V rms ±3% as measured with item MO2-1 at a frequency of 2.4 kHz ±.3%.
5. Adjust power supply PO1 until item MO2-2 indicates .15 V ±1%.
6. Measure the parallel inductance with the item LO1, for each test condition. Items MO2-1 and MO2-2 shall be removed after the voltages are set to reduce loading effects.

FIGURE 4. Inductance test circuit for table III - Continued.

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TABLE IV. DC resistance tests.

Test cond	Parameter	Connect		Initial limits 25° ±5°C amb		Units	Test fig
		A to pin	B to pin	Min	Max		
C1	L1 dc resistance	2	3	60.3	73.7	mΩ	3
C2	L2	6	7				
C3	L3	22	23				
C4	L4	14	15				
C5	L5 dc resistance	18	19	60.3	73.7	mΩ	3

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TABLE V. Insulation resistance tests.

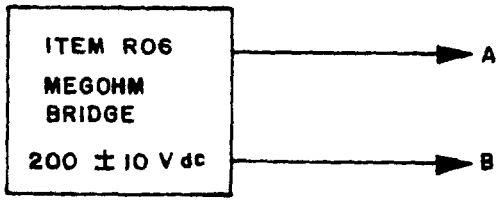
Test cond	Parameter	Connect		Initial limits 25° ±5°C amb		Units	Test fig
		A to pin	B to pin	Min	Max		
D1	L1 Insulation resistance	1	11,5, 13,17, 21	1000	∞	MΩ	5
D2	L2	5	1,11 13,17, 21				
D3	L3	13	1,5,11, 17,21				
D4	L4	17	1,5,11, 13,21				
D5	L5 Insulation resistance	21	1,5,11, 13,17	1000	∞	MΩ	5

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TABLE VI. Module integrity test.

TEST COND	PARAMETER	CONNECT		INITIAL LIMITS 25° ±5°C AMB		UNITS	TEST FIG
		A TO PIN	B TO PIN	MIN	MAX		
E	Module integrity	1-8,11, 13-24	9,10,12, 25-40	1000	∞	MΩ	5

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads.

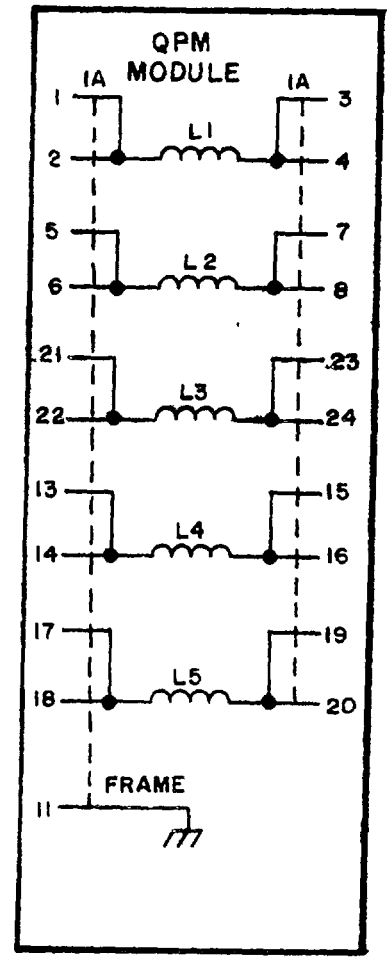


FIGURE 5. Insulation resistance test circuit for tables V and VI.

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TABLE VII. Frame continuity test.

Test cond	Parameter	Connect		Initial limits 25° ±5°C amb		Units	Test fig
		A to pin	B to pin	Min	Max		
F	Frame continuity	11	Fin	0.0	0.1	Ω	3

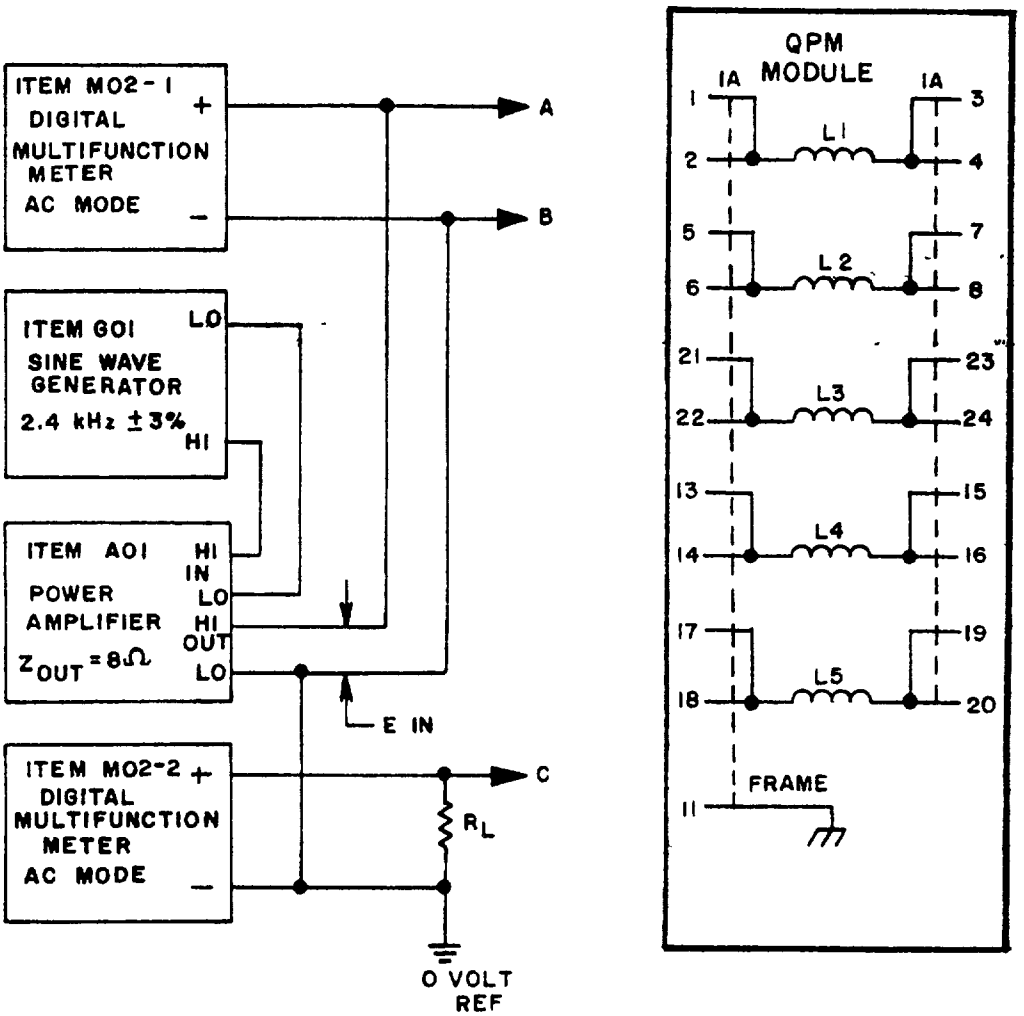
NOTE: With a sharp needle-like probe attached to the ohmmeter lead, pierce (do not scratch) the insulated surface coating of the module fin and measure the resistance between the fin and pin 11.

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TABLE VIII. Electromagnetic coupling tests.

Test cond	Parameter	Connect			Initial limits 25° ±5°C amb		Units	Test fig
		A to pin	B to pin	C to pin	Min	Max		
G1	L1-L2 Electro-magnetic coupling	1	3,7, 11	5	0.0	40	mV	6
G2	L1-L4	1	3,11, 15	13				
G3	L1-L5	1	3,11, 19	17				
G4	L1-L3	1	3,11, 23	21				
G5	L2-L4	5	7,11, 15	13				
G6	L2-L5	5	7,11, 19	17				
G7	L2-L3	5	7,11, 23	21				
G8	L4-L5	13	15,11, 19	17				
G9	L4-L3	13	15,11, 23	21				
G10	L5-L3 Electro-magnetic coupling	17	19,11, 23	21	0.0	40	mV	6

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NOTES:
1. R_L = 100Ω ±1%, 1/8 W.
2. E_{in} = 2.0 V rms ±1%.

FIGURE 6. Electromagnetic coupling test circuit for table VIII.

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TABLE IX. Jumper resistance tests.

TEST COND	PARAMETER	CONNECT A TO PIN	CONNECT B TO PIN	INITIAL LIMITS 25° ±5°C AMB MIN	INITIAL LIMITS 25° ±5°C AMB MAX	UNITS	TEST FIG
H1	L1 Jumper resistance	1	2	0.0	0.020	Ω	3
H2	L1	3	4				
H3	L2	5	6				
H4	L2	7	8				
H5	L3	21	22				
H6	L3	23	24				
H7	L4	13	14				
H8	L4	15	16				
H9	L5	17	18				
H10	L5 Jumper resistance	19	20	0.0	0.020	Ω	3

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